

Module 1 Question Pool

OMSCS 6250 Computer Networks

Lesson 1: Introduction, History, and Internet Architecture

Layering & Architecture

Q1. [MCQ]

A company replaces the Ethernet links in its office network with Wi-Fi. The hosts run the same applications and the same TCP/IP software as before. The applications keep working after the change.

Which property of layering explains why the applications keep working?

- A. A layer's design can change, as long as the layer still offers the same service to the layer above.
- B. Each layer reads the headers of the layers below it, so the applications adjust to the new link.
- C. Each layer serves the layer below it, so the new Wi-Fi link adapts itself to the applications above.
- D. The transport layer rewrites each frame for the new link, so the application data stays the same.

Correct answer: **A**

Why: The move from Ethernet to Wi-Fi changes only the link layer. The link layer still offers IP the same service, so every layer above it runs unchanged. Each layer serves the layer above it and reads only its own header. Frames are built by the data link layer, and the transport layer never handles them.

Q2. [TF]

Two ISPs use different link technologies. One ISP uses fiber, and the other ISP uses a cellular network. Neither ISP changes its link technology to match the other ISP. The customers of the two ISPs exchange IP traffic with each other.

The customers can exchange IP traffic because each link technology below IP provides the service that IP expects.

- True
- False

Correct answer: **True**

Why: IP runs on every host and router, and it works over any link technology that provides the service IP expects. Fiber and cellular both provide that service, so traffic crosses from one ISP to the other with no change to either link. A lower layer can differ, or change, as long as it still provides the same service upward.

Per-Layer Roles

Q3. [TF]

A developer writes an application that sends data over TCP.

The application passes its data to the transport layer by writing the data into the payload of an IP datagram.

- True
- False

Correct answer: **False**

Why: The application hands its data to the transport layer through the socket interface. The transport layer puts the data in a segment, and the network layer builds the IP datagram around that segment.

Q4. [MCQ]

A web browser process on host A sends data to a web server process on host B. The data crosses several routers between the two hosts.

Which layer of the Internet protocol stack delivers the data to the web server process on host B?

- A. Application layer
- B. Data link layer
- C. Network layer
- D. Transport layer

Correct answer: **D**

Why: The transport layer delivers data from process to process. It adds port numbers, so the data reaches the right process on host B, here the web server. The network layer carries the packet only as far as host B, and the data link layer carries a frame across one link at a time. The application creates the data and relies on the transport layer to deliver it.

Q5. [MCQ]

A developer must choose between TCP and UDP as the transport protocol for a new application.

Which statement correctly compares the services that TCP and UDP provide to the application?

- A. TCP provides reliable, in-order delivery and congestion control; UDP provides best-effort delivery.
- B. TCP provides reliable, in-order delivery and congestion control; UDP provides the same reliability.
- C. TCP provides reliable delivery but no congestion control; UDP provides congestion control but no reliability.
- D. TCP is connectionless but provides reliable delivery; UDP is connection-oriented but offers best-effort delivery.

Correct answer: **A**

Why: TCP is connection-oriented. It delivers data reliably and in order, and it runs flow control and congestion control. UDP is connectionless and best-effort: it sends each datagram once and leaves reliability and rate control to the application.

Q6. [TF]

At a source host, the transport layer passes a segment down to the network layer. The network layer creates a datagram from the segment.

The datagram is routed across multiple networks based on the destination IP address.

- True
- False

Correct answer: **True**

Why: The network layer adds its header to each segment to form a datagram. That header carries the source and destination IP addresses. Each router forwards the datagram hop-by-hop using the destination IP address, and no fixed end-to-end path is set up beforehand.

Q7. [MCQ]

A datagram travels from a source host to a destination host through three routers. At each hop, the network layer passes the datagram down to the data link layer.

What does the data link layer do with the datagram at each hop?

- **A.** It provides end-to-end reliability between the application processes on the two hosts.
- **B.** It moves the datagram wrapped inside a frame across one link to the next node on the path.
- **C.** It chooses the route toward the destination network using the datagram's IP address.
- **D.** It signals each raw bit over the copper, fiber or radio medium of the link.

Correct answer: **B**

Why: The data link layer carries a frame across one link, from one node to the next. Choosing the route belongs to the network layer, signaling bits to the physical layer, and end-to-end reliability to the transport layer.

Q8. [TF]

A source host sends a datagram to a destination host. The datagram crosses three networks on the way.

The data link layer is responsible for moving the datagram from the source host all the way to the destination host.

- True
- False

Correct answer: **False**

Why: The data link layer moves a frame across one link. The network layer moves the datagram across all the links between the hosts.

Encapsulation

Q9. [TF]

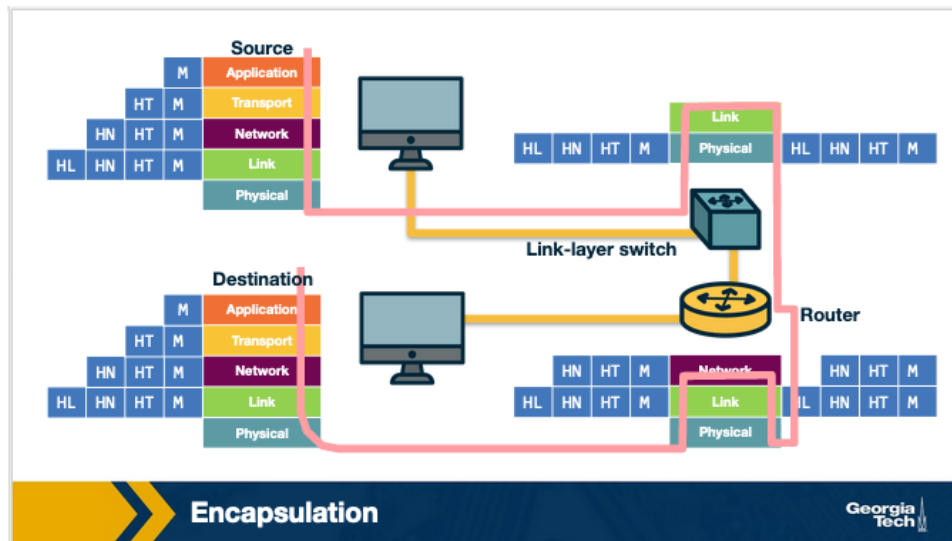


Figure: Layered encapsulation across the protocol stack (Module 1)

A host sends a message to a server in another network. At the host, the transport layer, the network layer and the data link layer each add a header.

At the server, the data link layer removes its header first, and the transport layer removes its header last.

- True
- False

Correct answer: **True**

Why: The receiver undoes encapsulation in reverse order. The data link layer removes the link-layer header first. The network layer then removes the network-layer header. The transport layer removes the transport-layer header last and passes the message up to the application.

Q10. [MCQ]

An application passes its data to the transport layer at the sending host. The data moves down the Internet protocol stack. The transport layer, the network layer and the data link layer each add a header. The unit of data has a different name at each layer.

What is the unit of data called at the network layer?

- A. Segment
- B. Frame

- C. Datagram
- D. Message

Correct answer: **C**

Why: Each layer has its own name for the unit it sends: a message at the application layer, a segment at the transport layer, a datagram at the network layer, a frame at the data link layer, and bits at the physical layer.

Q11. [MCQ]

A host sends a packet to a server in a different network. A router between the host and the server forwards the packet toward the server.

Which layers of the protocol stack does the router implement to forward the packet?

- A. Network and transport layers, reading IP addresses and ports.
- B. Physical and data link layers, reading MAC addresses.
- C. All five layers, the same stack the two end hosts run.
- D. Physical, data link and network layers, reading IP addresses.

Correct answer: **D**

Why: A router receives bits, reads the frame on each link, and forwards on the IP address, so it implements the physical, data link and network layers. Ports belong to the transport layer, which only the end hosts run, along with the application layer. A switch stops at the data link layer.

End-to-End Principle

Q12. [MCQ]

Saltzer, Reed, and Clark asked where a network should place functions such as reliable delivery. Their answer is the end-to-end principle.

Which statement describes the core claim of the end-to-end principle?

- A. Functions like reliable delivery work correctly only with help from the endpoints, so the core stays simple.
- B. Each function belongs in the lowest layer that can do it, so the endpoints have less work left to do.
- C. The core should provide reliable delivery, and the endpoints should add the functions the core lacks.
- D. Endpoints should keep their connection state in the routers, so a host crash does not end the connection.

Correct answer: **A**

Why: Some functions, such as reliable delivery, work correctly only with help from the endpoints. Even if the core delivered reliably, the endpoints would still have to check that the data arrived intact. The principle therefore keeps the core simple and moves such

functions up to the endpoints, the opposite of placing each one in the lowest layer that can do it. Fate-sharing keeps connection state in the hosts, where it is lost only when a host fails.

Q13. [TF]

A file transfer application needs every byte to arrive, in order. A video call application needs low delay and tolerates some loss.

According to the end-to-end principle, the network core should give both applications the same reliable, in-order delivery.

- True
- False

Correct answer: **False**

Why: A simple core lets each application build the service it needs at the edges: the file transfer adds reliability and ordering, and the video call leaves them out to keep delay low. One universal behavior in the core would limit what applications could do.

Violations of End-to-End: NAT and Firewalls

Q14. [MCQ]

A NAT-enabled home router translates between a private home network and the public Internet. A device inside the home network sends a packet to a server on the Internet.

Which fields of the packet does the NAT typically rewrite?

- **A.** The destination IP address, and usually the destination port number too.
- **B.** The source IP address, and usually the source port number too.
- **C.** The source and destination IP addresses, and usually both port numbers too.
- **D.** The source port number, while the private source IP address stays as it is.

Correct answer: **B**

Why: On the way out, the NAT replaces the private source address with its public address, and often the source port as well. The destination stays as the host wrote it. Keeping the private address would make the reply unroutable.

Q15. [TF]

A laptop connects to the Internet through a home NAT. The laptop has the private IP address 10.0.0.4. No port forwarding is configured on the NAT. The translation table of the NAT holds no entry for the laptop. A server on the public Internet tries to open a connection to the laptop.

The connection attempt from the server fails.

- True

- False

Correct answer: **True**

Why: The server can reach only the NAT's public address. With no mapping, the NAT has no inside host to deliver to. Applications that need inbound connections use workarounds such as STUN or UDP hole punching.

Q16. [MCQ]

A campus network has a firewall at its border. A home router runs NAT for a private home network.

Why are both the firewall and the NAT considered violations of the end-to-end principle?

- **A.** Both read the headers of the packets in transit before they forward the packets toward the destination.
- **B.** Both exist to save public IPv4 addresses, and the end-to-end principle forbids sharing an address.
- **C.** Both sit between the endpoints and change the communication, filtering traffic or rewriting addresses.
- **D.** Both drop packets when their buffers fill up, so delivery between the two endpoints is not guaranteed.

Correct answer: **C**

Why: A firewall decides which traffic may pass, and a NAT rewrites addresses and ports. Both sit in the path between the endpoints and interfere with the communication. Routers also read headers, and they drop packets when their buffers are full, yet routers do not violate the principle. NAT exists to save IPv4 addresses; firewalls exist for security.

Violations of End-to-End: NAT and Fate-Sharing

Q17. [MCQ]

A laptop connects to the Internet through a home NAT router. The laptop is downloading a large file from a server over a TCP connection. The router reboots and is back up 30 seconds later, with an empty translation table. The laptop and the server keep running the whole time.

What happens to the download?

- **A.** It resumes, because TCP retransmits lost segments and both endpoints still hold their connection state.
- **B.** It fails, because the mapping that sent the server's packets to the laptop was lost with the router.
- **C.** It continues, because the server still sends to the router's public address, which did not change.
- **D.** It fails, because the router held the connection's sequence and acknowledgment counts, now gone.

Correct answer: **B**

Why: The server's packets arrive at the router's public address and port, and the router needs its translation table to rewrite them toward the laptop's private address. The reboot erased that table, so the packets have nowhere to go and the connection breaks, although neither endpoint failed. The public address alone cannot identify the inside host. TCP retransmission recovers from the

reboot of an ordinary router, which holds no connection state. The sequence and acknowledgment counts live in the endpoints; a NAT breaks fate-sharing because it also holds state the conversation depends on.

Q18. [MCQ]

An engineer designs NAT-2, a NAT with a main box and a standby box. The main box copies every entry of its translation table to the standby box. If the main box fails, the standby box takes over with the same mappings. The designers of the Internet wanted the state of a conversation to be lost only when an endpoint that holds the state is lost.

Which statement correctly evaluates NAT-2 against this design goal?

- A. It meets the goal, because the state now survives unless both boxes fail.
- B. It falls short only because the copies to the standby box can lag.
- C. It falls short, because the state still sits in the middle of the network.
- D. It meets the goal, because the extra box makes the connection more reliable.

Correct answer: **C**

Why: The goal ties the conversation's state to the endpoints: the state may be lost only when an endpoint is lost. NAT-2 still keeps the state in the middle of the network, now in two boxes. If both boxes fail, the connection ends while both endpoints are healthy. Extra reliability and faster copying change how often the state is lost, but the goal concerns where the state lives.

Hourglass & Ossification

Q19. [MCQ]

The Internet protocol stack has an hourglass shape. There are many link technologies at the bottom and many applications at the top. A small set of core protocols sits in the middle, which is the waist of the hourglass.

Which statement describes how quickly each part of the stack has changed?

- A. The waist changes often, while the link technologies and the applications change slowly.
- B. The applications change often, while the link technologies and the waist change slowly.
- C. The link technologies and the waist change often, while the applications change slowly.
- D. The link technologies and the applications change often, while the waist changes slowly.

Correct answer: **D**

Why: Ethernet, Wi-Fi, fiber and cellular all appeared and spread, and new applications keep arriving. IPv4, TCP and UDP have been far more stable, because so much depends on them. Applications change fast because they need no change in the waist.

Q20. [TF]

The hourglass shape of the Internet protocol stack has a cost called ossification.

Ossification means that new application-layer technologies struggle to spread.

- True
- False

Correct answer: **False**

Why: Ossification means the core protocols resist change even when better ones exist. Applications above the waist still change quickly, and application-layer experimentation remains fast.

Evolutionary Architecture Model (EvoArch)

Q21. [MCQ]

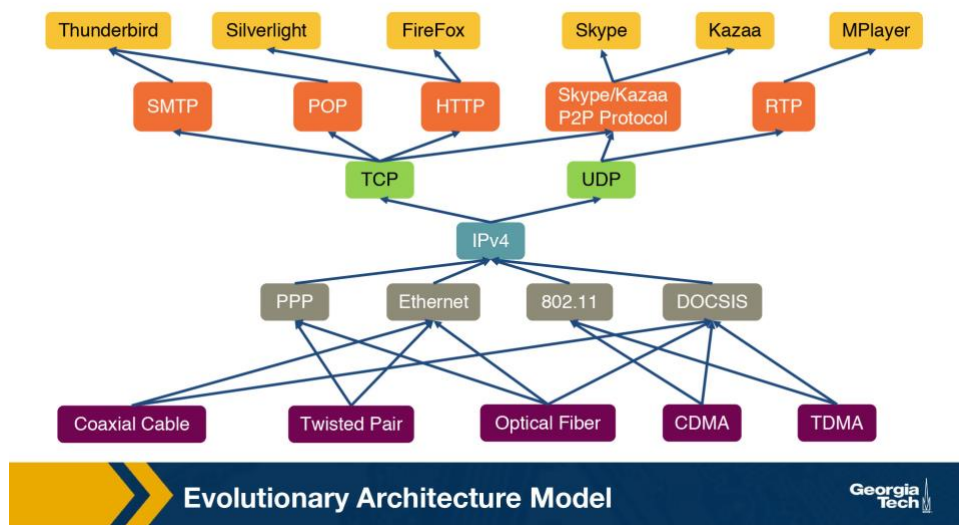


Figure: EvoArch layered protocol dependency graph (Module 1)

EvoArch models a protocol stack as a layered graph. Each node is a protocol. An edge joins two protocols in adjacent layers when the higher protocol uses the lower one. For example, TCP runs over IPv4, and HTTP runs over TCP.

Which statement defines the substrates and the products of a protocol?

- **A.** A protocol's substrates are the lower-layer protocols it uses; its products are the higher-layer protocols that use it.
- **B.** A protocol's substrates are the higher-layer protocols that use it; its products are the lower-layer protocols it uses.
- **C.** Substrates and products both name the horizontal competitors that operate at the very same layer of the stack.
- **D.** A protocol's substrates are the older protocols it was built to replace; its products are the newer ones replacing it.

Correct answer: **A**

Why: A protocol's substrates are the protocols one layer below that it uses, and its products are the protocols one layer above that use it. Both terms describe how protocols in adjacent layers depend on each other, whatever their age. Protocols at the same layer can compete, and they are neither substrates nor products of each other.

Q22. [TF]

In EvoArch, every protocol has an evolutionary value. TCP runs over IPv4, and many applications run over TCP.

The evolutionary value of TCP is the sum of the values of its substrates, the protocols beneath TCP.

- True
- False

Correct answer: **False**

Why: EvoArch sums the values of a protocol's products, the protocols that use it. The protocols beneath TCP add nothing to its value. TCP's value is high because many applications depend on it.

Q23. [MCQ]

In an EvoArch run, protocols u and w are at the same layer. Each of the two protocols uses some protocols in the layer below and is used by some protocols in the layer above.

According to EvoArch, when does w compete with u?

- **A.** When one protocol is defined as the successor to the other in a later revision of the same standard.
- **B.** When they share enough of the same products, the higher-layer protocols that use both of them.
- **C.** When they share enough of the same substrates, the lower-layer protocols that both of them run over.
- **D.** When one offers a higher-quality service than the other, so that users would move to the better one.

Correct answer: **B**

Why: Two protocols compete when they share enough of the same products, the higher-layer protocols that use them. EvoArch decides competition from shared products alone, so shared substrates, better quality or a successor standard never make two protocols rivals.

Q24. [MCQ]

A research group designs NextTransport, a new transport protocol. In every workload that the group measured, NextTransport is faster and more efficient than TCP.

According to EvoArch, why might NextTransport still fail to gain wide adoption on the Internet?

- **A.** Because NextTransport would share no products with TCP, and protocols without shared products are the first to die.

- **B.** Because the services already built on TCP give it high evolutionary value, and that incumbency protects it.
- **C.** Because TCP has more substrates beneath it, and a protocol's value comes from the protocols it runs over.
- **D.** Because a better protocol spreads only after a standards body retires the older one, as none has for TCP.

Correct answer: **B**

Why: A protocol's value is the sum of its products' values. The many services built on TCP give it high value, while NextTransport starts with little and loses when it competes for those products. A protocol that shared no products with TCP would not compete with it at all, and could coexist. Substrates add no value, and EvoArch leaves standards bodies out of the model.

Q25. [MCQ]

EvoArch gives each layer l a parameter $s(l)$. A new protocol at layer $l+1$ selects each protocol at layer l as a substrate with probability $s(l)$.

What does the parameter $s(l)$ capture?

- **A.** It is measured at runtime from traffic volume, so a layer carrying more bytes is assigned a higher value.
- **B.** It is a policy that standards bodies apply to keep the number of protocols at each layer roughly in balance.
- **C.** It describes how general a layer's service is, higher at the lower layers that serve many products.
- **D.** It describes how reliable a layer's protocols are, higher at the layers whose protocols rarely fail.

Correct answer: **C**

Why: The parameter $s(l)$ describes how general the service of layer l is. Lower layers are more general, because more kinds of protocols above can use them: a layer-1 protocol moves bits between two connected points, while SMTP serves only email applications. $s(l)$ is fixed before a run, independent of traffic, reliability or any standards body.

Q26. [MCQ]

A team designs NetX, a new network-layer protocol. IPv4 is the evolutionary kernel at the network layer. The team wants NetX to survive.

According to EvoArch, which launch plan gives NetX the best chance of surviving?

- **A.** Offer services that IPv4 lacks, and move TCP and UDP onto NetX so that it takes IPv4's place.
- **B.** Offer the same services as IPv4 with a faster design, so that its higher quality raises it above IPv4.
- **C.** Offer the same services as IPv4 over more link types, so it gains more substrates than IPv4 has.
- **D.** Offer services that IPv4 lacks, and deploy NetX only for the applications that need those services.

Correct answer: **D**

Why: Protocols compete when they share products. A protocol that offers services IPv4 lacks gains products of its own, so it can grow without competing with IPv4. Moving TCP and UDP onto NetX would make it share IPv4's products while its value is still lower,

and IPv4 would win, as it did against IPv6. A faster version of the same service is still a direct competitor, and more substrates add no value, because value comes from products.

Q27. [MCQ]

An EvoArch run has many layers. Layer X, near the bottom, has a generality probability of 0.9. Layer Y, in the middle, has a generality probability of 0.5. Layer Z, near the top, has a generality probability of 0.1. New protocols are born evenly across the layers.

In which layer does EvoArch predict that protocols are most likely to die?

- A. Layer Y, because products per node vary most there, so a few nodes far outweigh their rivals.
- B. Layer X, because nodes there have the most competitors, so each faces the fiercest competition.
- C. Layer Z, because nodes there have the fewest products, so each has low value and loses to rivals.
- D. No single layer, because births are spread evenly, so deaths are spread evenly as well.

Correct answer: **A**

Why: The number of products a node receives is a binomial draw with probability $s(l)$, and its spread is widest near 0.5. At layer Y a few nodes collect far more products, and far more value, than their neighbors; they compete with most of their layer and win, so death peaks there. At layer X nodes share nearly the same products, so their values are similar and death is rare. At layer Z nodes have so few products that they rarely share enough to compete. Deaths follow competition, which differs by layer, so even births do not give even deaths.

Q28. [MCQ]

In EvoArch, IPv4 is the evolutionary kernel at the network layer. EvoArch describes TCP and UDP as an evolutionary shield for IPv4.

Why do TCP and UDP protect IPv4 from being replaced?

- A. IPv4 survives because it runs over increasingly diverse link-layer technologies, which add to its value.
- B. IPv4 survives because the steady stream of new applications keeps adding value to TCP and UDP, and so to IPv4.
- C. IPv4 keeps its value and survives, since few new transport protocols last long enough to pick a rival network layer.
- D. IPv4 gains value and survives as its features grow, since EvoArch adds value to a protocol for its capabilities.

Correct answer: **C**

Why: A kernel is threatened by new protocols in the layer above it. For IPv4 that layer is the transport layer, where new protocols rarely survive competition with TCP and UDP, so the rivals that could weaken IPv4 never last. New applications barely change the value of TCP and UDP, because the two already have so many products. A protocol's value comes from the protocols that use it, so the link technologies below IPv4 and IPv4's own features add nothing.

Interconnecting Hosts and Networks

Q29. [MCQ]

A small network has one hub and one Layer-2 switch. Hosts P and Q are connected to the hub. A third port on the hub is connected to the switch. Hosts R and S are connected to the switch. The forwarding table of the switch already maps every host to the correct port. Host P sends a frame addressed to host R.

Which hosts, other than host P, receive a copy of the frame?

- **A.** R alone, because the hub reads the destination and repeats the bits only toward the switch.
- **B.** Q and R, because the hub repeats the bits to every other port and the switch sends only toward R.
- **C.** Q, R and S, because the hub and the switch each send the frame out of all their other ports.
- **D.** Q alone, because the switch sees the frame came in from a hub port and keeps it on that side.

Correct answer: **B**

Why: A hub repeats the bits that arrive on one port out of all its other ports, so Q receives the frame, and P and Q share one collision domain. The switch reads the destination MAC address, finds R in its table, and sends the frame out of R's port only, so S does not receive it. Choosing one output port by destination is a switch's job, and a hub reads no addresses. A switch forwards by destination address, whichever port the frame came in on.

Q30. [MCQ]

An office network connects 20 hosts to one hub. When only a few hosts send traffic, the network performs well. When more hosts send traffic at the same time, throughput falls and delays grow. The office then replaces the hub with a Layer-2 switch. The hosts and their traffic stay the same, and performance recovers.

Which statement explains the difference in performance between the hub and the switch?

- **A.** The hub drops frames whenever its buffer fills, while a switch stores them until the busy link is free.
- **B.** The hosts on the hub share one collision domain; a switch sends each frame only toward its destination.
- **C.** The hub reads IP addresses, which is slow, while a switch reads MAC addresses, which is faster.
- **D.** The hub's host table overflows as more hosts send, so it falls back to sending each frame to all ports.

Correct answer: **B**

Why: A hub repeats every bit out of all its other ports, so all 20 hosts compete for one shared medium, and collisions grow with traffic. A switch forwards each frame only toward the port that leads to its destination MAC address, so the hosts stop competing for one medium. A hub reads no addresses and keeps no table or buffer. Buffering belongs to the switch, which stores frames while an output link is busy.

Learning Bridges

Q31. [MCQ]

A learning bridge maintains a forwarding table that maps MAC addresses to ports. A frame arrives at the bridge. The destination MAC address of the frame is already in the forwarding table.

What does the bridge do with the frame?

- **A.** It forwards the frame on the port recorded for the frame's source address, where that host was last seen.
- **B.** It floods the frame on every port except the one it arrived on, to cover a possibly stale table entry.
- **C.** It forwards the frame only on the port that the table associates with that destination address.
- **D.** It rewrites the destination MAC to the port's own address before forwarding the frame onward.

Correct answer: **C**

Why: A known destination means the bridge forwards the frame only on the learned port, which saves bandwidth on the other segments. The bridge floods only frames whose destination is not in its table. The source address tells the bridge where the sender is, and the frame leaves with its MAC addresses unchanged.

Q32. [MCQ]

A learning bridge is redesigned so that it drops any frame whose destination MAC address is not in its forwarding table, instead of flooding the frame. The forwarding table starts empty. Host A sends a frame to host B. Host B has not sent any frame yet.

What happens to the frame from host A when the frame arrives at the bridge?

- **A.** It is dropped, because the bridge has no table entry for host B's address.
- **B.** It is forwarded, because the bridge records host B's address from the frame.
- **C.** It is held until host B sends a frame, and then it is forwarded to host B.
- **D.** It is forwarded, because the bridge now finds host A's entry in its table.

Correct answer: **A**

Why: The bridge looks up the destination, host B. Host B has not sent a frame, so the table has no entry for it, and this redesigned bridge drops the frame. A bridge learns only from source addresses: it records host A from this frame, and that entry serves later frames addressed to host A. A bridge forwards, floods or drops each frame as it arrives, and it holds no frame waiting for another host to send.

Q33. [MCQ]

A learning bridge connects three hosts. Host A is on port 1, host B is on port 2 and host C is on port 3. The forwarding table starts empty. Three frames cross the bridge in this order: host A sends a frame to host B, host B sends a frame to host A, and then host C sends a frame to host B.

Which hosts receive a copy of the third frame, from host C to host B?

- A. Hosts A and B, because host C is new to the bridge, so the bridge floods the frame.
- B. Hosts A and B, because the bridge learns only from frames it floods, so it knows only host A.
- C. Host B only, because the bridge learned host B's port when host B sent the second frame.
- D. Host B only, because the bridge learned host B's port from the destination of the first frame.

Correct answer: **C**

Why: The bridge learns from the source address and arrival port of every frame. The first frame teaches it that host A is on port 1; host B is still unknown, so that frame is flooded to ports 2 and 3. The second frame teaches it that host B is on port 2. When host C's frame arrives, host B is in the table, so the bridge sends it only on port 2. The bridge also records host C on port 3, and a new source never causes a flood; only an unknown destination does. The bridge learns where hosts are from source addresses only.

Spanning Tree

Q34. [TF]

Network designers add redundant links between bridges to improve reliability. The redundant links create a loop in the Layer-2 topology. The bridges run no loop-prevention protocol.

A single broadcast frame keeps circulating around the loop indefinitely.

- True
- False

Correct answer: **True**

Why: Layer-2 frames carry no hop limit like the TTL field of IP packets, so nothing removes a frame that goes around a loop. The bridges keep forwarding the same broadcast frame again and again, and the network becomes congested.

Q35. [MCQ]

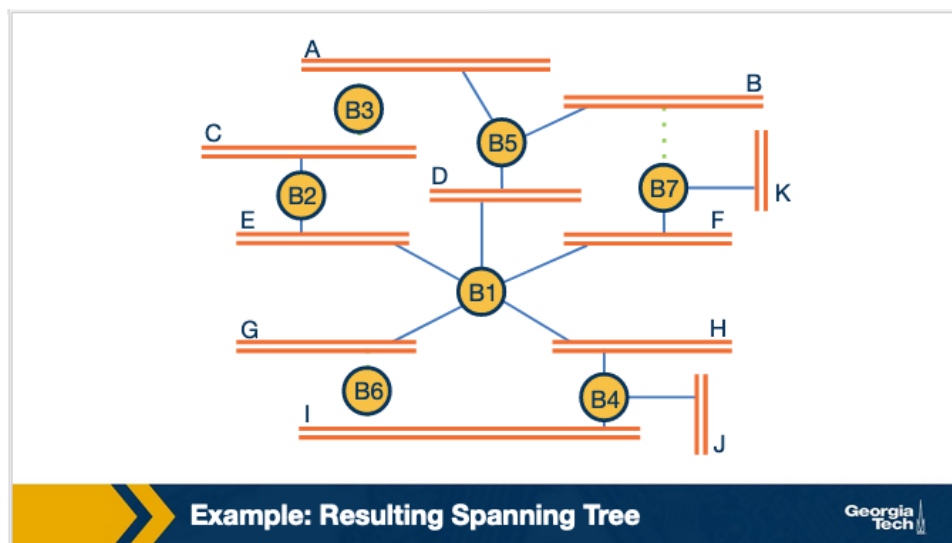


Figure: Resulting spanning tree (Module 1)

A campus network connects its bridges with redundant links. The redundant links create loops in the Layer-2 topology. The bridges run the Spanning Tree Algorithm.

What does the Spanning Tree Algorithm do in this network?

- **A.** It raises link capacity by combining the redundant links into a single high-bandwidth virtual link.
- **B.** It picks a root bridge that relays the traffic between the other bridges, so that no frame takes a long path.
- **C.** It deletes the redundant links from the topology, so that the remaining links form a tree with no loops.
- **D.** It blocks forwarding on some ports so frames cannot loop, while every physical link stays in place.

Correct answer: **D**

Why: The algorithm blocks forwarding on some ports, so the active topology has no loops. The blocked links stay connected and can carry traffic again if a link fails. The root bridge is the reference point from which the tree is built, and frames between any two bridges travel along the tree. Combining links to raise capacity is a different technique.